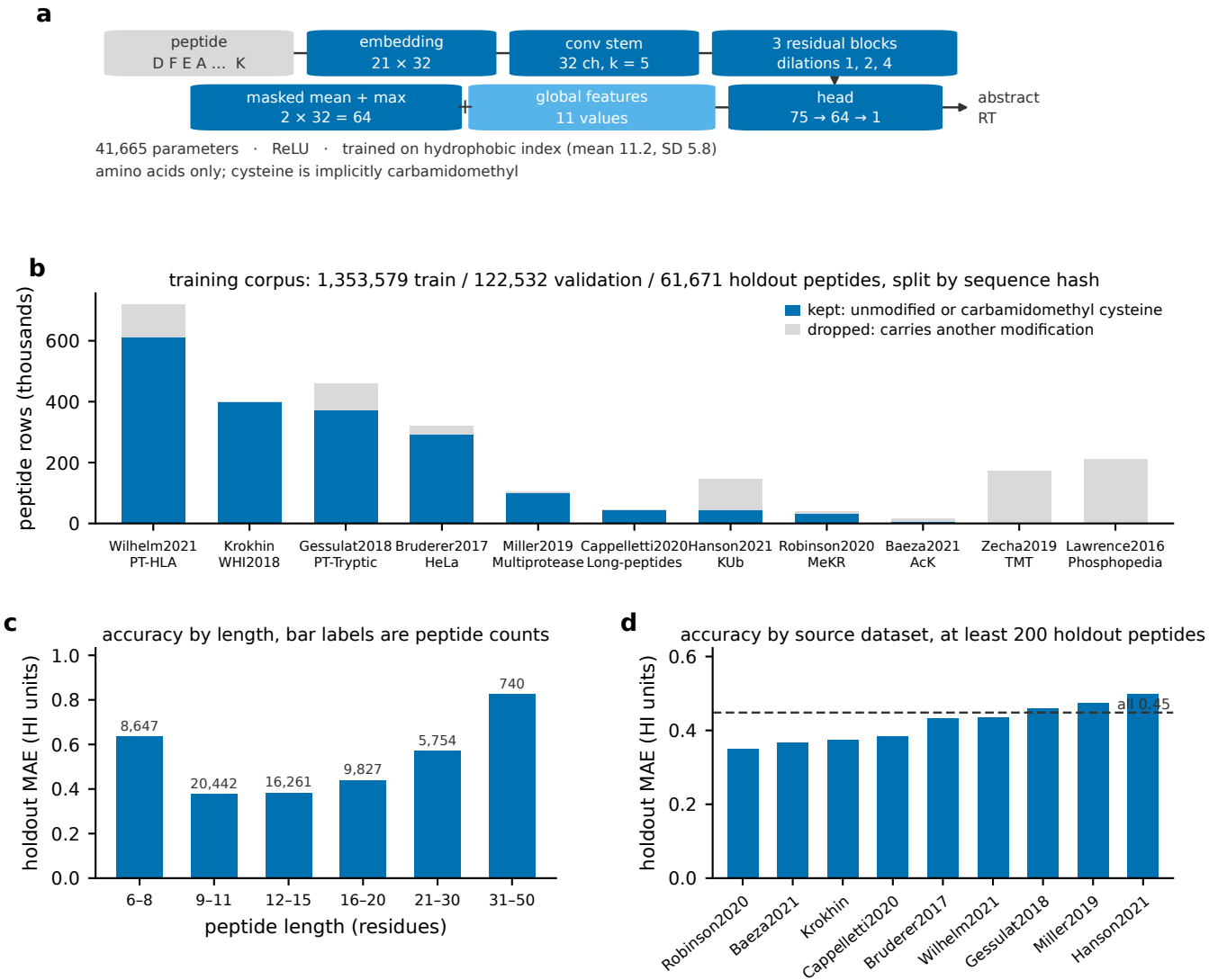

a
peptide
D F E A … K
embedding
21 × 32
conv stem
32 ch, k = 5
3 residual blocks
dilations 1, 2, 4
masked mean + max
2 × 32 = 64
global features
11 values
head
75 → 64 → 1
abstract
RT
41,665 parameters · ReLU · trained on hydrophobic index (mean 11.2, SD 5.8)
amino acids only; cysteine is implicitly carbamidomethyl
b
training corpus: 1,353,579 train / 122,532 validation / 61,671 holdout peptides, split by sequence hash
kept: unmodified or carbamidomethyl cysteine
dropped: carries another modification
peptide rows (thousands)
0
200
400
600
Wilhelm2021 PT-HLA
Krokhin WHI2018
Gessulat2018 PT-Tryptic
Bruderer2017 HeLa
Miller2019 Multiprotease
Cappelletti2020 Long-peptides
Hanson2021 KUb
Robinson2020 MeKR
Baeza2021 AcK
Zecha2019 TMT
Lawrence2016 Phosphopedia
c
accuracy by length, bar labels are peptide counts
holdout MAE (HI units)
0.0
0.2
0.4
0.6
0.8
1.0
8,647
20,442
16,261
9,827
5,754
740
6–8
9–11
12–15
16–20
21–30
31–50
peptide length (residues)
d
accuracy by source dataset, at least 200 holdout peptides
holdout MAE (HI units)
0.0
0.2
0.4
0.6
all 0.45
Robinson2020
Baeza2021
Krokhin
Cappelletti2020
Bruderer2017
Wilhelm2021
Gessulat2018
Miller2019
Hanson2021